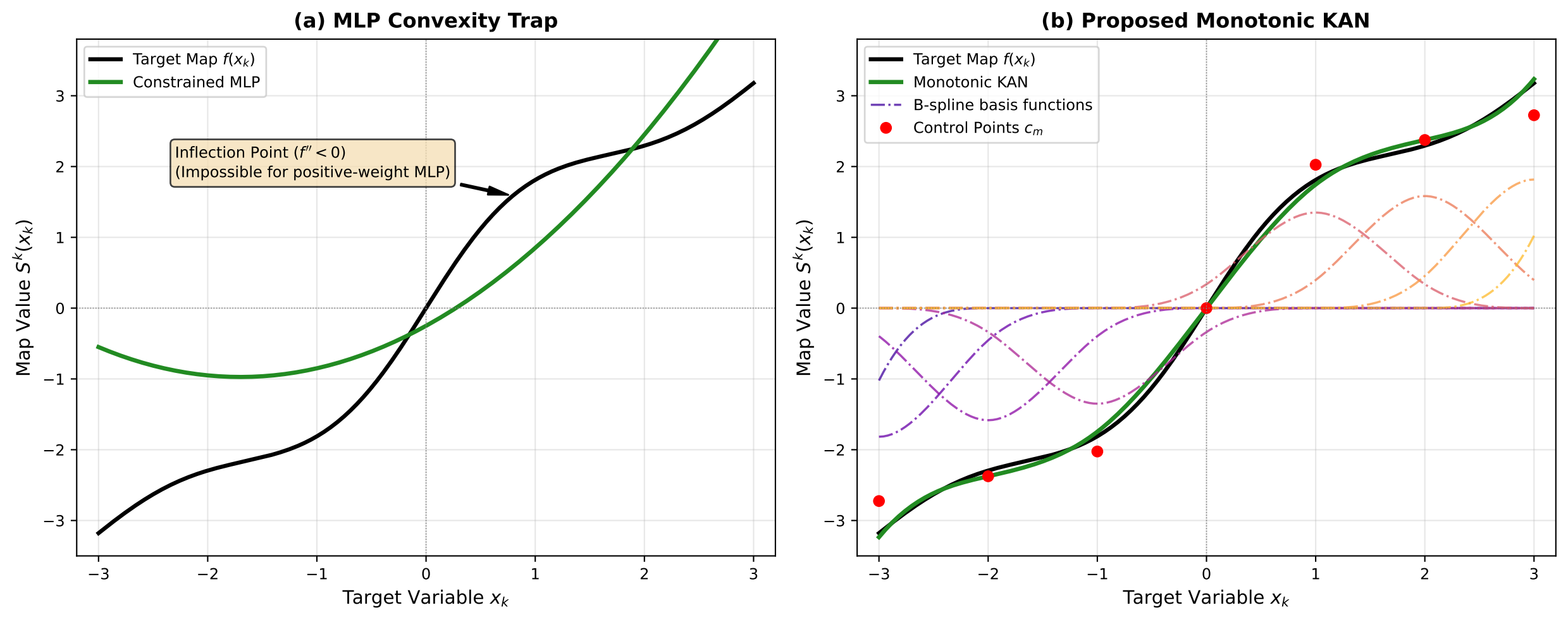
(a) MLP Convexity Trap
Target Map $f(x_k)$
Constrained MLP
Inflection Point ($f'' < 0$)
(Impossible for positive-weight MLP)
Map Value $S^k(x_k)$
Target Variable $x_k$
(b) Proposed Monotonic KAN
Target Map $f(x_k)$
Monotonic KAN
B-spline basis functions
Control Points $c_m$
Map Value $S^k(x_k)$
Target Variable $x_k$